Math 105L **Lab 4 Part 2**

Human Growth: Rates of Change, Acceleration, & Concavity

1. Background: Zoomed Data on Ages 8–14 Years (95.5 to 160.5 Months)

In this lab session, we examine the relationship between the graph of stature (height $H(t)$ in cm) and the graph of rate of growth ($H'(t)$ in cm/year):

- **Height function $H(t)$:** Represents the median stature of children at age t (in months).
- **Growth rate $H'(t)$:** Computed as $\frac{\Delta H}{\Delta t} = \frac{H(t+\Delta t)-H(t)}{1/12 \text{ yr}}$, giving growth speed in cm/year.
- **Growth Acceleration $H''(t)$:** The rate of change of the growth rate, measured in cm/year^2.

The Core Mathematical Connections: Shape of $H(t)$ vs. Behavior of $H'(t)$

- **Growth Spurt Peaks (Maximum of $H'(t)$):**
  - When the rate curve reaches a local maximum, the child is growing at their fastest possible rate.
  - On the height graph $H(t)$, this corresponds to the point of **maximum steepness**.
  - This point is an **inflection point** of the height curve where concavity switches from up to down.
- **Growth Rate Valleys (Minimum of $H'(t)$):**
  - When the rate curve reaches a local minimum, the growth has slowed to its minimum rate before the adolescent growth spurt begins.
  - On the height graph $H(t)$, this corresponds to the point of **least steepness**.
  - This point is also an **inflection point** where concavity switches from down to up.

2. Male vs. Female Growth Comparison

From the classroom data projection:

- **Female Adolescent Growth Spurt Peak:** Occurs at approximately **140 months** ($\approx$ 11.7 years). At this age, the female height curve is steepest, after which growth decelerates and the curve bends downward (concave down) toward adult stature.
- **Male Adolescent Growth Spurt Peak:** Occurs later, at approximately **160+ months** ($\approx$ 13.3+ years). Male peak growth rate is higher and occurs roughly 1.5–2 years later than the female peak.
- **Pre-Spurt Minima (Troughs):** Occur at $\approx$ 115 months for females and $\approx$ 125 months for males.

3. Acceleration and Concavity Formalization

- **Acceleration > 0 ($H''(t) > 0$):** The growth rate $H'(t)$ is increasing. The height graph is **concave up** (bending upward).
- **Acceleration < 0 ($H''(t) < 0$):** The growth rate $H'(t)$ is decreasing. The height graph is **concave down** (bending downward).
- **Inflection Point of $H(t)$:** Occurs wherever acceleration passes through zero ($H''(t) = 0$ with a sign change), corresponding exactly to the peaks and troughs of the rate curve!